The Journal of Open Source Software

solids4foam-v2.1: A toolbox for performing solid mechanics and fluid-solid interaction simulations in OpenFOAM

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 01 July 2024
Published: unpublished

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Summary

`solids4foam` is a toolbox designed for conducting solid mechanics and fluid-solid interaction simulations within the widely-used OpenFOAM software (ESI-OpenCFD, 2024; foam-extend, 2024; Foundation, 2024). The toolbox has a comprehensive set of features, including advanced algorithms for fluid-solid and thermo-fluid-solid coupling, a variety of solid material models, non-trivial solid boundary conditions, and numerous discretisation and solution methods for solid mechanics.

Statement of Need

The `solids4foam` toolbox addresses four primary needs within the OpenFOAM community:

1. The need to perform fluid-solid interactions using OpenFOAM.
2. The need to solve complex solid mechanics problems directly within OpenFOAM.
3. The necessity for a modular approach to coupling various solid and fluid processes in OpenFOAM.
4. The demand for an extendable framework to facilitate research into innovative finite volume methods for solid mechanics.

The design of `solids4foam` adheres to four guiding principles:

1. **Usability:** If you can use OpenFOAM, you can use `solids4foam`.
2. **Compatibility:** Supports the three main OpenFOAM forks: OpenFOAM.com, OpenFOAM.org, and foam-extend.
3. **Ease of Installation:** The toolbox is easy to install and requires minimal additional dependencies beyond OpenFOAM.
4. **Code Quality:** Emphasis on code design and style, closely following the OpenFOAM coding style guide.

Features

`solids4foam` employs a modular design, offering generic class interfaces for solid mechanics, fluid dynamics, fluid-solid coupling methods, and solid material models. It also supports all native OpenFOAM modularity, including boundary conditions and function objects.

The `solids4foam-v2.1` release includes the following features:

Partitioned Fluid-Solid Interaction Coupling Methods

- Fixed under-relaxation (Tuković, Karač, et al., 2018)
- Aitkens accelerated under-relaxation (Tuković, Karač, et al., 2018)
- Interface-quasi-Newton coupling (Degroote J, 2009)
- Robin-Neumann coupling (Tuković, Bukač, et al., 2018)
- Thermo-fluid-solid interaction coupling

Finite Volume Solid Model Discretizations and Solution Algorithms

- Segregated (Cardiff et al., 2018), coupled (Cardiff, Tuković, Jasak, et al., 2016), and explicit solution algorithms
- Linear geometry (small strain) and nonlinear geometry (finite strain) formulations, including total and updated Lagrangian
- Cell-centered and vertex-centered formulations
- Continuum and plate formulations

Solid Material Models

- Linear elasticity (isotropic, orthotropic (Cardiff et al., 2014)), plasticity (J_2 (Cardiff, Tuković, De Jaeger, et al., 2016), Mohr-Coulomb (Tang et al., 2015)), viscoelasticity (Cardiff et al., 2018), thermo-elasticity (Cardiff et al., 2018), poroelasticity (Tang et al., 2015)
- Hyperelasticity (neo-Hookean, Ogden, Mooney-Rivlin (Oliveira et al., 2022, 2023), Fung (Oliveira et al., 2022, 2023), Yeoh (Oliveira et al., 2022, 2023)), hyperelastoplasticity (Cardiff, Tuković, De Jaeger, et al., 2016)
- Interface to Abaqus material model subroutines (UMATs)

Solid Boundary Conditions

- Frictional contact (node-to-segment (Cardiff et al., 2012; Cardiff, Tuković, De Jaeger, et al., 2016), segment-to-segment (Batistić et al., 2022, 2023))
- Cohesive zone models
- Traction, displacement, rotation

Fluid Models

- Incompressible (PIMPLE, PIMPLE-overset)
- Multiphase (volume-of-fluid)
- Weakly compressible (Oliveira et al., 2022)

Function Objects

- Energies, displacements, forces, stresses, principal stresses, torques

Utilities and Scripts

- Scripts for ensuring compatibility with the main OpenFOAM forks
- Mesh conversion utilities: OpenFOAM to/from Abaqus

Tutorials

- A suite of example cases and benchmark problems to demonstrate functionality and verify performance

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Acknowledgements

The solids4foam toolbox has not directly received funding from any source; nonetheless, many implementations within the toolbox have stemmed from research projects at University College Dublin, the University of Zagreb, and other institutes. With this in mind, Philip Cardiff gratefully acknowledges financial support from the Irish Research Council through the Laureate program, grant number IRCLA/2017/45, and the European Research Council (ERC) under the European Union's Horizon 2020 Research and Innovation programme (Grant agreement No. 101088740), Science Foundation Ireland (SFI) through I-Form (SFI Grant Number 16/RC/3872), MaREI (SFI Grant Number RC2302 2), NexSys (SFI Grant Number 21/SPP/375), as well as the DJEI/DES/SFI/HEA Irish Centre for High-End Computing (ICHEC) for the provision of computational facilities and support (www.ichec.ie), and the UCD ResearchIT Sonic cluster funded by UCD IT Services and the UCD Research Office.

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